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Time-to-expiration for evidence bodies of medication harms (TIME): survival analysis from a retrospective cohort

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Abstract

Background Systematic reviews and meta-analyses are cornerstones of evidence-based medicine, yet their conclusions can be revised or overturned as new evidence emerges. Prior research has mapped how systematic-review conclusions evolve, but no empirical study has quantified how evidence on drug harms shifts over time. We aimed to examine the stability of safety evidence.

Methods We conducted a before-and-after comparison study using the SMART Safety dataset, a well-established empirical resource for evidence synthesis of medication-related harms. For each eligible meta-analysis, we aggregated trial-level 2×2 data annually and applied the IVhet model to generate cumulative effect estimates over time; Kaplan–Meier survival analysis was applied to estimate time to signals of evidence expiration and convergence. We defined signals of evidence expiration as the first change in effect direction, statistical significance, or a $\geq 50\%$ relative change in the pooled odds ratio (OR), and convergence as a ≤ 0.10 absolute difference between consecutive cumulative ORs over five annual iterations. Median times to expiration and convergence were then estimated across eligible meta-analyses.

Results We included 505 meta-analyses, spanning 310 harm outcomes, with a median of 328 adverse events per meta-analysis. During follow-up, signals of evidence expiration occurred in 356 (70.5%) of these, with a median time of 3 years (95% CI 3 to 4). Among expiration events, 117 (32.9%) were triggered solely by a $\geq 50\%$ relative change in effect magnitude, and 89 (25.0%) met two or three criteria simultaneously. Subgroup analyses showed similar patterns, with medians ranging from 2 to 7 years. For the 416 meta-analyses with $I^2 < 50\%$, 278 (66.8%) met the expiration definition, with a median of 4 years (95% CI 3 to 4). Only 27 meta-analyses (5.3%) met the convergence criterion during follow-up; the 5th and 25th percentiles for time to convergence were 9 and 31 years, respectively, while the median was not estimable.

Conclusions Current evidence suggests that meta-analytic evidence on harms may expire within a relatively short time, while signals of convergence tend to appear much later. These observations point to cautious interpretation of meta-analytic evidence on harms and suggest that timely updating should be considered.

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Keywords Medication harms · Evidence expiration · Evidence convergence · Survival analysis

Background

Systematic reviews and meta-analyses serve as core methodologies in evidence-based medicine [1, 2]. By synthesizing individual study findings, they provide clinicians, guideline developers, and policymakers with generally more credible evidence summaries [1]. Such evidence, however, is not always dependable, particularly when relevant clinical trials are insufficient for a given clinical question [3]. In this case, as further clinical evidence emerges, earlier conclusions may be modified or overturned. Recognizing this practical constraint, Cochrane moved from updating reviews whenever new evidence appeared to a two-year interval and eventually to a priority-based updating model [4].

The Cochrane policy on review updating stemmed from a 2007 survival analysis of 100 Cochrane reviews [5]. For each review, the team searched for new trials, updated the synthesis, and compared the new results with the original ones in terms of changes in statistical significance, direction of effect, or a 50% change in effect magnitude. The analysis revealed that 23% of these reviews exhibited an update signal within two years and 50% within 5.5 years as new trials accumulated. This study provides a valuable paradigm for understanding how clinical evidence evolves, which is particularly important for informed healthcare decision-making.

However, these findings may not apply to the evolution of evidence concerning medication-related harms, which differ substantially from efficacy endpoints in statistical and clinical respects [6, 7]. Since adverse events are generally rare, relevant meta-analyses frequently suffer from low statistical power and considerable random error [8, 9]. In addition, inconsistent definitions and measurement methods introduce systematic error into trial-derived safety estimates [10]. These flaws collectively weaken the robustness of meta-analytic estimates for adverse events, rendering the corresponding evidence more fragile. Currently, no empirical study has examined how evidence on medication-related adverse events evolves over time. Given that medication-related harms may cause permanent injury, prompt withdrawal of approved treatments or other regulatory interventions, timely identification of evolving safety signals is critical for clinical practice and public health decision-making [11].

In this study, motivated by unresolved uncertainties around how safety evidence evolves over time, we performed a large empirical investigation to quantify changes in meta-analytic estimates via survival analysis. Our findings are expected to form an empirical basis for optimizing drug safety surveillance schemes.

Methods

Data resources

This study followed the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) reporting guideline, whenever possible [12]. We adopted a before-and-after design and utilized data from the SMART Safety dataset, a well-established empirical resource dedicated to evidence synthesis of medication-related harms [13–15]. Adverse events were defined as "any adverse medical events occurring in patients or subjects during clinical practice" [13, 16–18]. The database exclusively includes systematic reviews of randomized controlled trials (RCTs) that focus on adverse events of healthcare interventions. Only binary adverse event outcomes were eligible for inclusion in the database, encompassing 362 harm outcomes ranging from mild manifestations (e.g., nausea and vomiting) to severe, rare outcomes (e.g., heart failure and all-cause mortality). Overall, the dataset includes 151 systematic reviews with 629 meta-analyses, involving more than 2,300 RCTs [19].

The SMART Safety dataset was constructed using a rigorous, standardized methodology. Systematic reviews were originally retrieved from PubMed, with publication dates restricted between 1 January 2015 and 1 January 2020 [10, 13, 14, 20]. The search covered 93.8% of published systematic reviews focusing on medication safety

within the same period, ensuring considerable representativeness. Literature screening and data extraction were completed following a rigorous, pre-established protocol, including dual independent screening and two-stage data extraction with cross-verification against original trial materials (e.g., full texts, supplements, ClinicalTrials.gov, and manufacturer reports). Detailed information regarding the search strategy, study selection, and standardized double-extraction procedures has been comprehensively described in our prior methodological publications [10, 13, 14, 20, 21].

Data collection

The following structured data at three levels were used for analysis: systematic review, meta-analysis, and individual trial. At the systematic review level, variables included unique review identifiers and corresponding author information. At the meta-analysis level, we retrieved meta-analysis identifiers, targeted harm endpoints, and outcome classification. At the individual trial level, key variables comprised the first author, study identifier, subgroup identifier (an indicator to differentiate multiple intervention arms within a single trial), 2×2 contingency table data (event counts and sample sizes for intervention and control groups), and trial publication year.

Outcomes

We evaluated two complementary outcomes characterizing the temporal evolution of cumulative medication-related harm evidence: evidence expiration and evidence convergence. Evidence expiration represents a signal for systematic review updates, indicating that newly accumulated data alters the preceding cumulative evidence [5].

The primary outcome was evidence expiration, defined as the first occurrence of unstable estimates in cumulative meta-analyses. Specifically, we adopted the same metrics from the landmark 2007 study, according to which evidence expiration was deemed to occur when the cumulative pooled odds ratio (OR) met any of the following criteria [5]: 1) a reversal in the direction of the pooled effect size (i.e., $OR > 1$ to $OR < 1$, or vice versa); 2) a change in significance of the P value (i.e., $P < 0.05$ to > 0.05); or 3) a $\geq 50\%$ relative fluctuation in pooled OR compared with the preceding cumulative estimate [10, 22, 23]. This set of criteria can effectively prevent P-hacking and overestimation caused by minor fluctuations in effect sizes near 1 (e.g., changes from an OR of 0.99 to 1.01).

The secondary outcome was evidence convergence, defined as cumulative ORs remaining consistent throughout the observational period, with no occurrence of evidence expiration at any time point. Convergence was operationally defined as an absolute difference ≤ 0.10 between two consecutive cumulative ORs for five consecutive annual iterations, where an update was defined as each year in which at least one trial was incorporated into the cumulative meta-analysis [24]. Given the inherent arbitrariness of threshold-based definitions, we performed sensitivity analyses using alternative criteria requiring two, three, or four consecutive iterations (all with tolerable differences ≤ 0.10) to evaluate the robustness of the convergence findings under less stringent stability criteria.

Statistical analysis

The synthesis was performed in two hierarchical steps: 1) *Within-year aggregation*: Log ORs and their variances were first calculated for each RCT based on 2×2 tables of events and participants. Then, Log ORs from all RCTs published in the same year were combined using the inverse-variance heterogeneity (IVhet) model to yield an annual pooled OR with 95% confidence intervals (CIs) [25]. Each annual subgroup, therefore, served as the basic analytical unit for subsequent cumulative meta-analyses and survival analysis. 2) *Across-year cumulative synthesis*: these annual pooled estimates were then ordered and sequentially accumulated to perform cumulative meta-analysis, which reflects the temporal evolution of the overall effect estimate with annual incorporation of new evidence. These cumulative estimates of pooled ORs were then used to define time-to-event outcomes (evidence expiration and convergence), which were subsequently analyzed using survival analysis.

For trials with zero events, the 0.5 continuity correction method was applied to obtain the second-order approximation of the log odds ratio. This correction effectively resolves the computational issue of log odds ratio and its variance estimation when zero events occur in one or both study arms, as validated in our prior investigations [9, 14, 26]. For multi-arm RCTs sharing a common control group, only the intervention arm with the largest total sample size was retained to avoid duplication of participants.

We conducted time-to-event analyses based on the aforementioned cumulative meta-analytic estimates, adopting publication year as the time scale. The first annual pooled estimate was defined as the study starting point, with either evidence expiration or evidence convergence of the pooled effect serving as the corresponding survival event. The Kaplan–Meier method was applied to estimate survival and failure functions [5]. Survival time percentiles (5th, 25th, 50th, 75th, and 95th) with corresponding 95% CIs were estimated for the endpoints of evidence expiration and evidence convergence.

To explore whether the magnitude of accumulated evidence and the number of incorporated trials affect the stability of cumulative meta-analytic estimates, we performed subgroup analyses stratified by four predefined factors: 1) the proportion of the cumulative sample size attributable to the earliest evidence ($\leq$0.10, 0.10–0.20, 0.20–0.30, > 0.30); 2) the proportion of included trials attributable to the earliest evidence ($\leq$0.10, 0.10–0.20, 0.20–0.30, > 0.30); 3) the total number of included trials (2–5, 6–10, 11–20, > 20) and 4) the total cumulative sample size (<1,000; 1,000–4,999; 5,000–9,999; $\geq$ 10,000) in meta-analysis. Categories were defined based on empirical considerations to allow comparisons across meta-analyses with different scales of evidence accumulation [26]. Furthermore, to assess the impact of heterogeneity on cumulative evidence estimates, we carried out an extra subgroup analysis stratified by heterogeneity across annual pooled effect estimates. Heterogeneity strata followed the standard I^2 cut-offs from Higgins et al.: 0–25% (low), 25–50% (moderate), 50–75% (large), and 75–100% (substantial) [27]. We additionally performed a sensitivity analysis restricted to evidence bodies with low to moderate heterogeneity, defined as $I^2 < 50\%$ [27].

To assess the robustness of the evidence expiration definition and the potential influence of unstable significance-threshold crossings in cumulative meta-analysis, we additionally performed sensitivity analyses using alternative operational definitions of evidence expiration: analyses excluding the statistical-significance criterion, analyses excluding the $\geq$ 50% relative-change criterion, and analyses using only the $\geq$ 50% relative-change criterion in the definition of evidence expiration. All statistical analyses were performed using Stata/SE 17.0 (StataCorp, College Station, TX, USA). Two-sided hypothesis testing was used throughout analyses, and a *P* value < 0.05 was defined as statistically significant. Analysis codes are provided in Additional file 1: Appendix 5.

Results

Baseline characteristics

From the SMART Safety database, we included all 629 meta-analyses derived from 151 systematic reviews. After a series of data cleaning and preprocessing steps, 505 meta-analyses were included in the final analyses of evidence expiration and convergence (Fig. 1). These 505 meta-analyses contained a total of 5,056 trial-level records and covered 310 distinct adverse event outcomes.

Across these meta-analyses (Table 1), the number of included RCTs per meta-analysis ranged from 2 to 70, with a median of 11 (IQR: 7 to 20). The total sample size per meta-analysis had a median of 5931 (IQR: 3,097 to 13,461). The total number of adverse events per meta-analysis had a median of 328 (IQR: 76 to 1,096). The publication span of included trials, defined as the time interval between the earliest and latest trial years, had a median of 9 years (IQR: 6 to 12), while the number of distinct publication years was 6 (IQR: 4 to 9), corresponding to a median of 5 cumulative updates (IQR: 3 to 8) beyond the initial evidence.

The earliest accumulated evidence, representing the starting point of each cumulative evidence, typically accounted for a small proportion of the overall cumulative evidence. The median sample size in the first publication

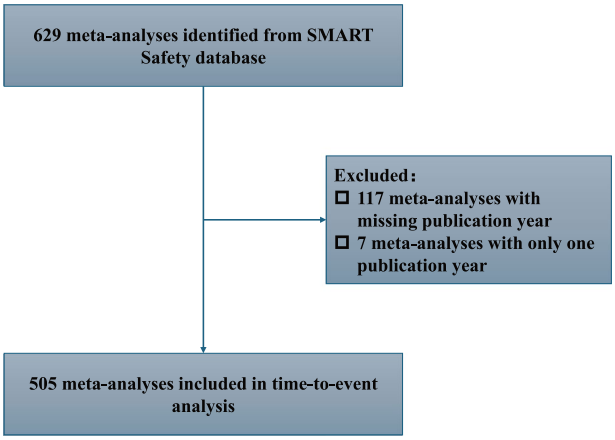


Fig. 1 Flowchart for screening eligible randomized controlled trials

Table 1 Baseline characteristics of included meta-analyses

Characteristic	Median (IQR) / n (%)
Total number of RCTs per meta-analysis	11 (7 - 20)
Total events per meta-analysis	328 (76 - 1096)
Total participants per meta-analysis	5931 (3097 - 13461)
Publication span (years)	9 (6 - 12)
Cumulative updates beyond first year	5 (3 - 8)
Participants in the first year	428 (160 - 1112)
Proportion of sample size in the first year	0.08 (0.02 - 0.20)
Number of RCTs in the first year	1 (1 - 2)
Proportion of RCTs in the first year	0.11 (0.06 - 0.20)
Statistical heterogeneity	
$I^2 < 25\%$	374 (74.1%)
$25\% \leq I^2 < 50\%$	42 (8.3%)
$50\% \leq I^2 < 75\%$	48 (9.5%)
$I^2 \geq 75\%$	41 (8.1%)

year was 428 (IQR: 160 to 1,112), accounting for 8% (IQR: 2% to 20%) of the total cumulative sample size. In terms of study count, the earliest-year evidence typically included one RCT (IQR: 1 to 2), representing 11% (IQR: 6% to 20%) of all included RCTs within each meta-analysis. Regarding statistical heterogeneity, most meta-analyses demonstrated relatively low between-study heterogeneity, with 374 meta-analyses (74.1%) having I^2 values between 0% and 25%.

Survival analysis: time to evidence expiration

Among 505 meta-analyses, 356 (70.5%) presented signals of evidence expiration during follow-up, while 149 (29.5%) remained stable by the end of observation. Of the expiration events, 92 (25.8%) were triggered exclusively by a change in statistical significance, 58 (16.3%) exclusively by a reversal in effect direction, and 117 (32.9%) exclusively by a $\geq 50\%$ relative change in effect magnitude, and the remaining 89 (25.0%) met two or three criteria simultaneously (Additional file 1: Fig S1).

Figure 2 presents the Kaplan–Meier analysis: the estimated time-to-expiration values at the 5th, 25th, 50th, 75th, and 95th percentiles were 1 (95% CI: 1 to 1), 1 (95% CI: 1 to 2), 3 (95% CI: 3 to 4), 11 (95% CI: 9 to 13), and 39 (95% CI: 28 to not estimable) years, respectively. Overall, half of the meta-analyses expired within 3 years, indicating rapid degradation in the reliability of early cumulative evidence of medication-related harms.

Subgroup analyses stratified by the proportional size of early evidence and the scale of the overall evidence showed consistent expiration patterns across all subgroups (Fig. 3). Median expiration times ranged from 2–4

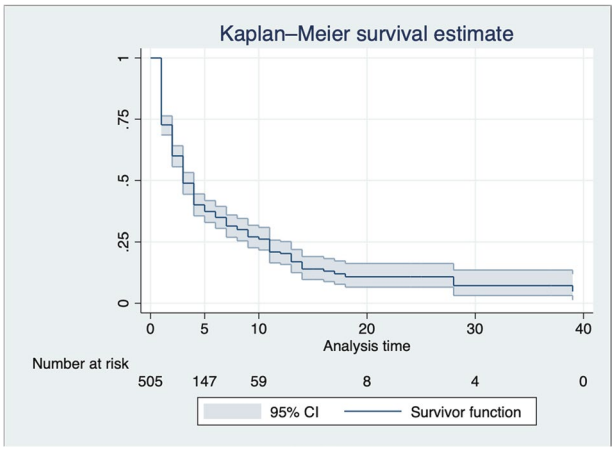


Fig. 2 Kaplan–Meier survival curve for time to evidence expiration

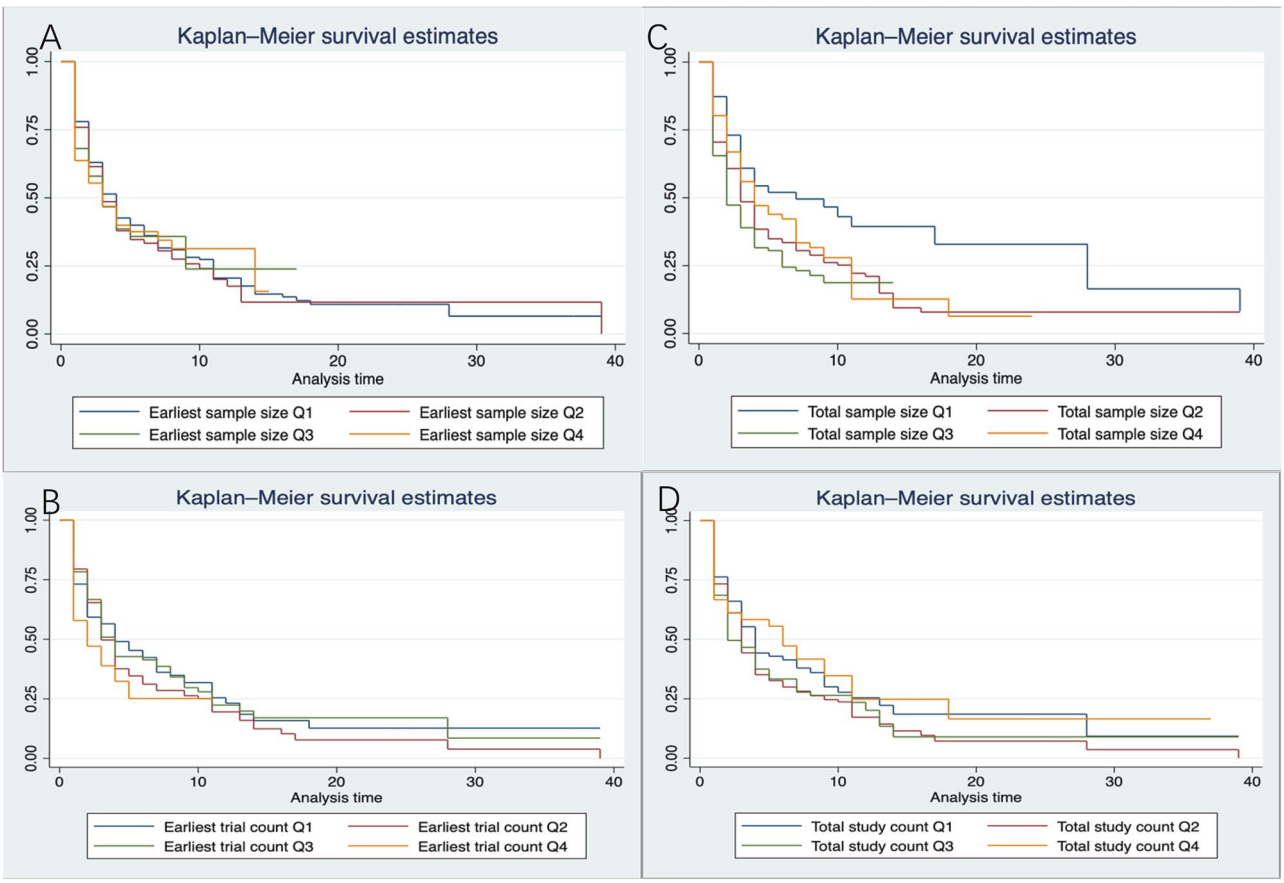


Fig. 3 Subgroup analyses of time to evidence expiration based on Kaplan–Meier survival curves. **A**–**B** show subgroup results based on (**A**) the proportion of first-year sample size and (**B**) the number of first-year RCTs; (**C**–**D**) show subgroup results based on (**C**) total cumulative sample size and (**D**) total number of included RCTs

years across the predefined categories of initial sample size proportion and initial RCT proportion. Similarly, stratification by total cumulative sample size and total number of included RCTs yielded median expiration times ranging from 2 to 7 years, with no clear dose–response gradient.

Subgroup analyses stratified by heterogeneity yielded varying median times to expiration (range 2 to 7 years, Fig. 4). The median time to expiration was 4 years (95% CI: 3 to 5), 3 years (95% CI: 2 to 4), 2 years (95% CI: 1 to 3), and 2 years (95% CI: 1 to 4) for meta-analyses with low to substantial heterogeneity, respectively.

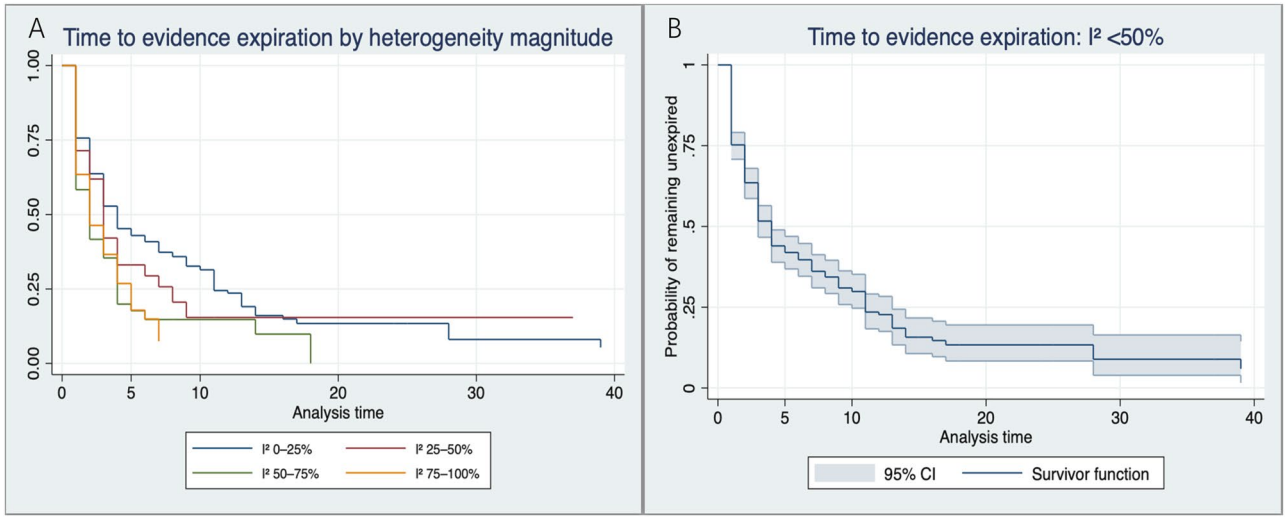
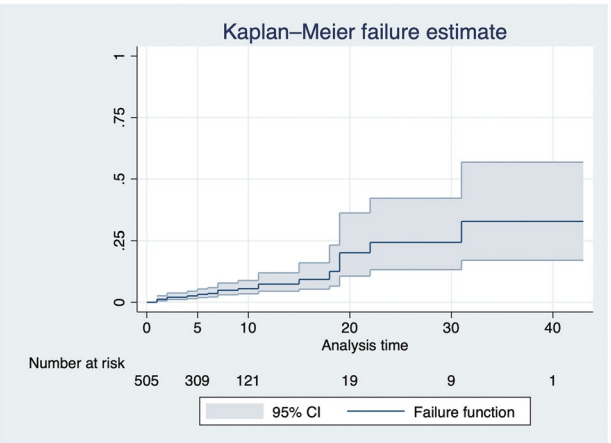


Fig. 4 Kaplan–Meier curves for time to evidence expiration stratified by heterogeneity. **A** Analysis stratified by I^2 category; (**B**) sensitivity analysis restricted to I^2 <50%

Fig. 5 Cumulative incidence curve for time to convergence



Expiration occurred earlier and more frequently in the higher-heterogeneity strata. In the sensitivity analysis restricted to meta-analyses with mild to moderate heterogeneity, 278 of 416 (66.8%) experienced expiration, with a median time to expiration of 4 years (95% CI: 3 to 4) (Additional file 1: Fig S2).

Survival analysis: time to evidence convergence

Only 27 (5.3%) of the 505 meta-analyses achieved predefined convergence during follow-up, whereas 478 (94.7%) remained unstable. In total, 341 (67.5%) meta-analyses never presented adjacent cumulative OR differences ≤ 0.10, and 137 (27.1%) exhibited only one to four eligible stable updates. The 5th and 25th percentile times to convergence were 9 years and 31 years, respectively, with no median convergence time estimable due to low overall stabilization rates (Fig. 5).

Subgroup analyses revealed distinct convergence patterns across evidence characteristics (Fig. 6). Convergence events were restricted to meta-analyses with the smallest initial sample size and fewest earliest trials, with negligible stabilization in other early-evidence subgroups. For the overall evidence scale, convergence exclusively occurred in large meta-analyses (>5,000 participants and ≥ 11 trials) with estimated convergence times of 19 to 22 years, while no convergence was observed in small-sample meta-analyses (≤1,000 participants). No

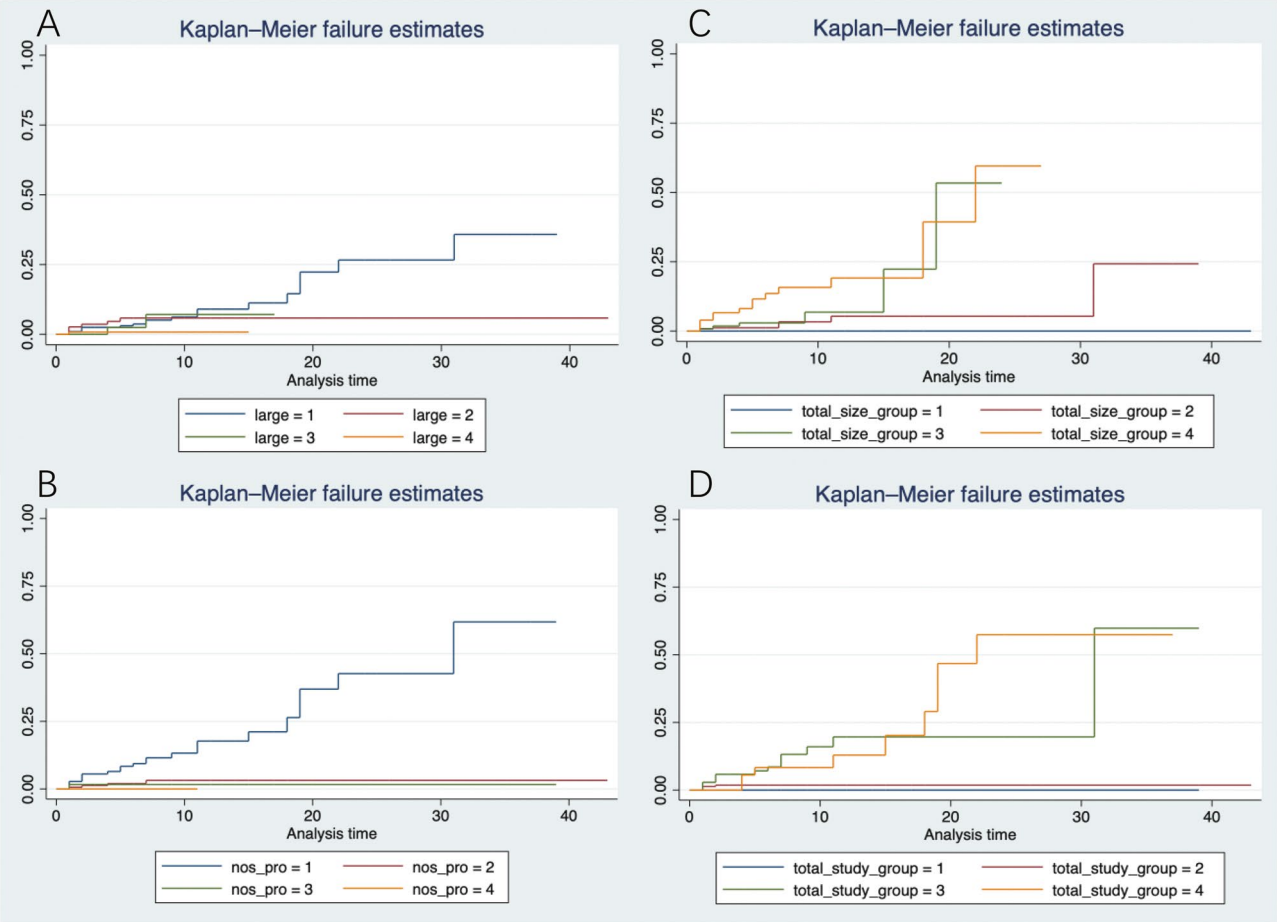


Fig. 6 Subgroup analyses of cumulative incidence curves for time to convergence. **A** Stratified by proportion of earliest sample size; (**B**) Stratified by proportion of earliest number of studies; (**C**) Stratified by total accumulated sample size; (**D**) Stratified by total number of included RCTs

subgroup reached a 50% cumulative convergence rate, so subgroup-specific median convergence times could not be estimated.

Post-hoc sensitivity analyses

Three modified definitions for evidence expiration yielded divergent expiration proportions (56.6%, 54.9%, 43.8%) and median expiration times (6, 5, 11 years) but consistent early-expiration patterns versus the primary analysis, with subgroup patterns unchanged across supplementary figures and tables. Alternative convergence criteria (2 to 4 consecutive updates) increased convergence rates and shortened stabilization durations (Additional file 1: Appendix 3); co-occurrence of expiration and convergence is summarized in Additional file 1: Appendix 4.

Discussion

In this study, we systematically characterized the temporal evolution and stability of drug adverse event evidence by analyzing 505 cumulative meta-analyses derived from published trial data. Our results demonstrated that evidence instability and expiration are extremely common in drug safety research and typically occur at the early stage of evidence accumulation. More than 70% of the early-stage evidence experienced substantive changes in

cumulative effect estimates, with half of the evidence expiring within three years. In contrast, sustained evidence convergence was exceedingly rare, achieved in only 5.3% of meta-analyses, and stable safety evidence usually required decades of continuous evidence follow-up. These findings align with our core hypothesis that meta-analyses of medication-related harms tend to be more fragile and become outdated more rapidly. They should also be interpreted with caution, as a substantial proportion of expiration events were triggered by criteria that may be susceptible to statistical noise and do not necessarily indicate clinically meaningful changes in safety evidence.

In our subgroup analyses, larger meta-analyses with greater sample sizes and more included trials tended to experience earlier evidence expiration, whereas small meta-analyses maintained abnormal stability for longer. This counterintuitive discrepancy is primarily driven by inherent statistical limitations of sparse safety data. Small meta-analyses often suffer from substantial random error, meaning a far larger amount of new evidence is required to overturn their initial pooled effect estimates, thereby prolonging the time until evidence expiration [28]. Furthermore, such false stability in small meta-analyses may also be due to the uneven speeds of evidence accumulation. Fast-growing research fields often experience rapid reversals in conclusions, whereas those with limited research activity exhibit spurious stability, a pattern that is described as the 'Proteus phenomenon' in epidemiology [29]. These findings further imply that for medication-related harms, even large meta-analyses do not necessarily ensure the reliability of the evidence.

Our findings have important implications for post-marketing monitoring and surveillance for adverse event. Currently, most medications are approved on the basis of two to three pre-approval trials, which constitute the earliest evidence for medication-related harms [30]. Our results demonstrate that half of this early evidence will become outdated within 3 years. This indicates that, in addition to real-world studies and routine adverse event monitoring systems, additional trials could be prioritized during the critical first 3 years post-marketing to further characterize and quantify the adverse event profiles of medications. Given that 95% evidence for medication-related harms expires within 39 years, suggesting that statistical changes may also occur considerably later and such comprehensive post-marketing monitoring may need long-term follow-up to ensure reliable adverse event profiles for most medications.

Regarding current update practices of systematic reviews, a two-year updating pattern may be insufficient for medication-related harms, as 25% of the evidence expires within the first year of evidence accumulation. This is reinforced by the fact that adverse events are typically designated as secondary rather than primary outcomes in both clinical trials and systematic reviews [31, 32]. On this basis, trials relevant to adverse event outcomes may be overlooked during the development and implementation of search strategies, which increases the uncertainty of evidence for harms. However, the real-world circumstances are far from what we can expect. As aforementioned, research activity differs across topics, with some drawing frequent new trials while others receive few updates for years. To this end, establishing living evidence platforms to capture the real-time evidence may represent a better alternative [33, 34].

Our study, using the largest safety dataset, provides cumulative evidence analyses that quantify time to expiration and convergence, revealing that safety evidence often becomes unstable early and far more frequently than previously recognized. These findings underscore the need for more responsive surveillance systems for drug safety, particularly in domains where adverse events are rare, inconsistently measured, and highly sensitive to evolving clinical practice [35]. Our outcomes also suggest that relying only on regular review updates may delay the detection of important shifts in harm data; integrating cumulative evidence monitoring into post-marketing assessment could help predict when new evidence is likely to meaningfully alter risk estimates [7].

Several limitations should be highlighted. First, our analyses were based on aggregated event counts rather than time-to-event data, leaving us unable to adjust for potential imbalances in treatment duration within and across trials. It has been estimated that a 20% difference in treatment duration between trial arms would result in biased estimation if unaccounted for [36]. Such bias may have some impact on the certainty of our findings. Second, the current study focused on meta-analyses of RCTs to capture the highest level of evidence for medication harms and inevitably neglected the value of real-world evidence. Real-world evidence can capture a broader range of safety signals in more heterogeneous populations and over longer time horizons and is therefore

considered complementary to RCTs in drug safety evaluation [37]. Third, the observational window may limit inference regarding long-term convergence. Because the median publication span within meta-analyses was 9 years, whereas the estimated 25th percentile of convergence was 31 years, some evidence may not have been followed long enough to demonstrate later convergence. As a result, the low observed frequency of convergence may partly reflect limited follow-up time rather than the true absence of long-term convergence.

Conclusion

Based on current best-available empirical evidence, our findings demonstrate that meta-analytic evidence on harms displays rapid and marked instability as new studies emerge, with most potentially expiring within a relatively short time and requiring prolonged accumulation of data to achieve convergence. These patterns suggest routine cumulative monitoring of harm data may help earlier identification of shifts and support more timely decision-making at the clinical and regulatory levels.

Abbreviations

IVhet	inverse-variance heterogeneity
CI	confidence interval
OR	odds ratio
RCT	randomized controlled trial

Supplementary information The online version contains supplementary material available at https://doi.org/10.1186/s12916-026-05296-8.

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Author contributions CX led the overall conception and design of the study. XW and CX performed the statistical analyses. XW organized the results and drafted the manuscript. CX, LFK, SAD, and EMW critically reviewed and revised the manuscript for important intellectual content. ZG, YL, SL, LL, SG, HC, and SV provided critical input during manuscript revision. XG reviewed and corrected the analytical code and contributed to the reanalysis of the revised results. All authors read and approved the final manuscript.

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Data availability Data can be found at https://osf.io/g3mdu/.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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